

Client Management & Intelligent Server Allocation System Functional & Technical Requirements Document

1. Project Overview

The system is designed to manage client accounts, algo subscriptions, intelligent server allocation, dealer mappings, and server capacity monitoring. Core objective: Automatically allocate the best possible server for a client while respecting all operational constraints.

2. Core Modules

- Authentication & Roles
- Master Client Management
- Algo-Server Mapping
- Intelligent Allocation Engine
- Capacity Monitoring
- Dealer Capacity Monitoring
- Reports & Dashboard
- Audit Logs
- Manual Override Panel

3. User Roles

Admin – Full access

Operator – Allocate/manage clients

Viewer – Read-only reports

Super Admin – Server creation/configuration

4. Updated Master Client Table

Table: clients_master

id – PK
user_id – Unique client ID
algo – Algo assigned
product – Product name
client_type – CC, MS, MSCC, Jainam, Achintya
broker – Broker name
dealer_id – Assigned dealer
status – Active / Inactive
pos_int – POS / INT
space_required – 1 or 2
allocated_server – Final assigned server
created_at – Timestamp
updated_at – Timestamp

5. Updated Algo-Server Mapping Table

Table: algo_server_mapping

server_name – Server identifier
algo – Supported algo
subscription – Subscription type
login – Login credential
total_capacity – Total server space
used_capacity – Used spaces
available_space – Remaining spaces
operator – Assigned operator
client_type_group – GROUP_A / GROUP_B
dealer_code – Dealer mapping
status – Active / Full / Disabled

6. Revised Grouping Logic

Group A

CC + MSCC

Group B

MS + Jainam + Achintya

Important Constraint:

Group A and Group B can NEVER share the same server.

7. Revised Server Capacity Rules

- Maximum clients per server = 27
- Maximum dealer clients = 20
- Space allocation = 1 or 2
- Group mixing is not allowed

User counts and dealer counts should be dynamically calculated from active clients instead of stored values.

8. Intelligent Allocation Engine

Step 1 — Determine Client Group

CC/MSCC → GROUP_A

MS/Jainam/Achintya → GROUP_B

Step 2 — Assign Dealer

Choose dealer with:

- Same group
- Active clients less than 20
- Prefer least utilized dealer

Step 3 — Find Eligible Servers

Server must satisfy:

- Algo Match
- Group Match
- Active Status
- Available space $\geq$ required space
- Active users < 27

Step 4 — Rank Servers

Priority 1 – Minimum remaining space

Priority 2 – Lowest active user count

Priority 3 – Balanced operator utilization

9. Allocation Failure Rules

Allocation should fail if:

- Dealer limit exceeded
- No eligible server found
- Invalid client grouping
- Server user limit reached
- Duplicate user ID detected

10. Manual Override

Admins should be able to:

- Force assign server
- Move users between servers
- Ignore optimization warnings
- Temporarily disable allocation rules

Hard limits must still be enforced and audit logs are mandatory.

11. Dashboard Requirements

Dashboard Widgets:

- Capacity Overview
- Active Servers
- Full Servers
- Available Capacity
- Client Type Distribution
- Dealer Utilization
- Server Heatmap

Heatmap Color Codes:

Green – Healthy

Yellow – Near Full

Red – Full

12. Important Edge Cases

Edge Case 1:

Server has user slots available but no space left → Reject allocation

Edge Case 2:

Dealer has 19 active clients → Allow allocation; after allocation dealer becomes full

Edge Case 3:

Server has enough user slots but insufficient space → Reject allocation

Edge Case 4:

Client becomes inactive → Free dealer slot and server space automatically

13. Final Recommended Database Design

Recommended tables:

- clients_master
- dealer_master
- server_master
- algo_server_mapping
- client_server_mapping
- allocation_logs
- audit_logs
- operators